

# TECHNOLOGY AT YOUR SERVICE

## 3D-printed shelter for astronauts on Mars

French company Fabulous has designed a conceptual shelter, dubbed Sfero, for future astronauts on planet Mars that would be 3D-printed on the red planet using locally-available materials such as the planet's own substrata. The process would involve a central mast



Resembling a futuristic igloo from the surface, 3D-printed shelter Sfero would be partially buried beneath the ground (Image courtesy: Fabulous)

that can extract iron from the planet's soil and rock, and also seek out permafrost to turn into water and use as insulation between the structure's inner and outer shell, reducing the effects of solar radiation.

The shelter resembling an igloo would be partially buried beneath the ground. Access to it would be gained by its one long corridor, which contains an airlock.

The interior would comprise three floors. The uppermost floor would measure 3sqm and food could be grown here, while the next floor down would measure 29sqm and comprise a work area and bathroom. The lowermost floor would measure 40sqm and contain sleeping quarters. Occupants would have to navigate between floors through a spiral staircase.

The project has been created in response to NASA's call for designs for 3D-printed habitats for Mars.

## 4D-printed self-folding structures

Researchers have developed a 4D printing technology using which complex self-folding structures could be created. They have used smart shape-memory polymers (SMPs) that can remember one shape and change to another programmed shape when uniform heat is applied.

The technology has been developed by the researchers at Georgia Institute of Technology, USA, and Singapore University of Technology and Design, Singapore, and could be used to create 3D structures that sequentially fold themselves from components that had been flat or rolled into a tube for shipment.

Ability to create objects that can change shape in a controlled sequence over time is enabled by printing multiple materials with different dynamic mechanical properties in prescribed patterns throughout the 3D object. When these components are heated, each SMP responds at a different rate to change its shape, depending on its internal clock.



The folded box is intended to simulate a postal mailer; folding is carried out in a tank of warm water to provide uniform heating to the materials (Image courtesy: Qi Laboratory)

This technology can have a range of applications; an unmanned air vehicle could change shape from one designed for a cruise mission to one designed for a dive.

## A flying four-seater hybrid-electric car

A start-up named Terrafugia, Massachusetts, USA, is working on a hybrid-electric four-seater flying car that can take-off and land vertically. According to the company, TF-X aircraft is designed to seat up to four people and will have a computer-controlled flight that will allow the operator to input the desired destination before letting the vehicle take-off and fly itself.

The flying car will have fold-out wings with twin electric motor pods attached to it and will be capable of recharging its batteries either from its engine or by plugging into electric car-charging stations.

A ducted fan will provide thrust, and motors will be powered by a 300hp engine and would move from vertical to horizontal positions as required for taking-off, cruising and landing. The vehicle will have a flight range of about 800km (500miles) with a cruising speed of about 322kmph (200mph).

TF-X will be capable of auto-landing at approved landing sites within approved weather limits, though the operator will have the final say over whether an approved landing zone is actually a safe place in which to land; the landing attempt can be aborted at any time.

The vehicle will also be capable of automatically